Orphans and Archetypes: Placing Unassigned Hittite Ritual Fragments in Their Traditions

Anonymous ACL submission

Abstract

Thousands of Hittite ritual and festival fragments cannot be assigned to a composition because no parallel passage survives. Laroche's residual category CTH 470 and the recently digitised EBo fragments hold 3,033 such manuscripts in the TLHdig 0.3 release, with a median of 10 legible words in CTH 470 and 4 among the EBo. We ask whether the words of a fragment and the way they are spelled can place it in a ritual or festival tradition, and how far such a placement can be trusted. Ten reference traditions are taken from the studies that argued for them and mapped to Laroche's CTH numbers. Nearest-centroid assignment over tf-idf features of word forms and sign sequences returns 84% of held-out manuscripts to their tradition (majority baseline 30%), and 57 of 78 compositions held out whole return to their scholarly group. A margin calibrated on held-out fragments of matching size lets the method abstain; it places 847 of 2,026 orphans at an expected precision of 80%. Splitting each decision exactly over its features shows that traditions are identified by what is written, above all agents, deities, realia and foreign-language words, and hardly by how it is spelled; common particles make up 47% of the similarity but 11% of the decision. Code and outputs are released, and every number in the paper is generated from them.

1 Introduction

Emmanuel Laroche's *Catalogue des textes hittites* (Laroche, 1971) gave every Hittite composition a number, and with it a place: rituals in one block, festivals in another, texts in Hattian, Luwian and Hurrian in blocks of their own. What could not be placed went into residual categories. CTH 470, "fragments of rituals", is the largest of them, and it has grown with every volume of hand copies since 1971. The Mainz Hethitologie-Portal keeps Laroche's numbers as a shared address system while moving fragments after joins (Košak et al., 2026; Košak, 2026). The digitisation of the EBo fragments in TLHdig 0.3 (Müller et al., 2026) adds 1,151 more manuscripts with a median of 4 legible words, most of them filed under two further residual numbers, CTH 832 and CTH 670.

A philologist assigns such a fragment through a parallel: a duplicate, or a close parallel passage in an identified composition. For most of these orphans that route is closed by definition. We ask instead whether the content of a fragment, its words and their spellings, can point to its tradition, and how far such a pointer can be trusted. Put this way, the task is attribution, and two known weaknesses of attribution apply with full force: accuracy falls with the length of the sample (Eder, 2015), and a classifier with a closed set of candidates always names one of them, even when the true source is not among them (Koppel et al., 2011). A placement is also only useful to a philologist who can check it, which requires knowing what it rests on.

Gordin et al. (2025) mapped about 1,000 Hurrian and Luwian festival manuscripts with tf-idf and t-SNE and proposed, as a next step, to test which word groups carry the signal. We turn that exploration into assignment, with explicit baselines, held-out evaluation and a rule for when not to assign. Our contributions are:

1. a reference of ten ritual and festival traditions, each defined by the study that argued for it and mapped to Laroche's CTH numbers, with the ranges that no study claims set aside as test sets (Section 3);

2. an evaluation that holds out manuscripts and whole compositions, so that the scholarly groupings themselves, and one contested attribution, are tested; that follows accuracy down to fragment size; and that calibrates a margin for abstention (Section 5);

3. an exact decomposition of each decision over word categories and writing layers, which separates what is written from how it is written and shows how much rests on common particles such as *nu* (Section 5.3);

4. a placement of 847 orphans, each with the features it rests on, as leads for verification on the tablet.

2 Background

Traditions and the catalogue. Laroche sorted by genre first and by language second, while later scholarship defines a ritual tradition by the origin of its practitioners, its cult and its ritual language. One tradition therefore ends up scattered over several CTH ranges. The Kizzuwatna rituals occupy CTH 471–500, but the ritual of the Kizzuwatnean Old Woman Maštigga is filed with the named practitioners at CTH 404 (Miller, 2004); the rituals of the AZU priest (CTH 701) sit in the Hurrian block (Salvini and Wegner, 1986); and the named-practitioner rituals of CTH 390–469 come from Arzawa, the Lower Land and Kizzuwatna alike (Christiansen, 2006; Bawanypeck, 2005). The festivals have been regrouped by cult geography (Gordin et al., 2025; Görke, 2022; Trameri, 2025) and, in the Hethitische Festrituale project (HFR), into edition modules such as the Luwian and the Palaic festivals (Rieken and Schwemer, 2026; Görke, 2020).

Scribal habit and attribution. Hittite scribes copied, excerpted and re-edited ritual texts over two centuries, and their habits of logographic writing, plene spelling and sign choice vary with date, school and individual (Gordin, 2015; Weeden, 2011; van den Hout, 2009; Waal, 2015). Whether a habit marks a tradition, a circle of scribes or a date is therefore a question to test, not an assumption. Stylometry supplies frequency-based representations, distance measures and protocols with held-out evaluation (Burrows, 2002; Juola, 2008; Stamatatos, 2009), and open-set recognition the means to reject a sample that fits no candidate (Scheirer et al., 2013).

Computational work on cuneiform. Machine learning on cuneiform has addressed transliteration and translation (Gordin et al., 2020; Yavasan and Gordin, 2025), language and dialect identification (Jauhiainen et al., 2019), the matching of fragments to known compositions by sign sequences (Simonjetz et al., 2024), and palaeographic style (Yugay et al., 2024). Fragment matching finds a known text; it does not place a fragment whose parallels are lost, and all of these studies are closed-set. We know of no published study that uses content features to assign Hittite fragments to religious traditions, or that asks when a fragment fits none of them.

3 Data

Corpus. We use the TLHdig Beta 0.3 release (Müller et al., 2026), converted to one row per word following Yavasan and Gordin (2025). Each word carries its normalised form, its transliteration split into signs, its language and, where the editors provide one, a morphological analysis. The release is downloaded by script and checked by its checksum, so the word table can be rebuilt from the public data alone. As in Gordin et al. (2025), reference manuscripts must have at least 17 legible words. The editors' analyses do not cover the whole release: the Luwian rituals (CTH 758–763) have none, and many other rituals lack them. The main model reads analysed words, so in its reference the Luwian rituals have 10 manuscripts and the western rituals 11; Section 5.2 repeats the evaluation on spellings alone with every manuscript.

Orphans. Orphans are all CTH 470 manuscripts and all EBo manuscripts, wherever TLHdig files them, 3,033 in all. CTH 670 and 832, where most EBo fragments are filed, are pools of orphans, not traditions. Of the 2,899 orphans in the CTH ranges we load, 2,026 have at least five legible words and are placed (median 7 words). Orphans never enter the reference.

Reference traditions. Each tradition is defined by the study that argued for it (Table 1). The Hurrian festival groups follow Table 3 of Gordin et al. (2025). Its single Luwian group is split as HFR splits it, into the festivals of Ištanuwa and Lallupiya and those of Ḫuwaššanna, and HFR's Palaic festivals are added (Rieken and Schwemer, 2026). The ritual traditions are the Kizzuwatna rituals as delimited by Miller (2004); the western rituals, that is, the augur rituals (Bawanypeck, 2005) and those whose colophon names Arzawa or Ḫapalla as the practitioner's home; the Luwian rituals of Puriyanni, Kuwattalla and Šilalluḫi (CTH 758–763), which Yakubovich and Mouton (2023) place in the Lower Land rather than in Kizzuwatna; and the Hattian cult layer of Klinger (1996), whose texts we take from the citations of his study in the Mainz Konkordanz

Tradition	CTH	Defining studies	n
Kizzuwatna and Lower Land (Hurr.)	628, 699–700, 711–713, 719, 722	Gordin et al. (2025) Table 3; Wegner and Salvini (1991)	318
Syro-Mesopotamia (Hurr.)	698, 701, 714–715, 721	Gordin et al. (2025) Table 3; Salvini and Wegner (1986)	167
Mixed (Hurr., Syr., Luw.)	704–706, 720	Gordin et al. (2025) Table 3	200
Ištanuwa and Lallupiya festivals	665, 771–773	Rieken and Schwemer (2026); Görke (2020); Hutter (2003)	52
Ḫuwaššanna festivals	690–694	Rieken and Schwemer (2026); Hutter (2003)	189
Palaic festivals	750–754	Rieken and Schwemer (2026); Görke (2020)	34
Kizzuwatna rituals	404, 471–499	Miller (2004); Strauß (2006); Görke (2010)	165
Arzawa (western) rituals	394, 402, 406, 410, 425, 433	Bawanypeck (2005); colophons	11
Luwian rituals (Lower Land)	758–763	Yakubovich and Mouton (2023)	10
Hattian cult layer	591, 627, 725–729, 731–746, 820–821	Klinger (1996)	480

Table 1: Reference traditions, the studies that define them, and the number of reference manuscripts with 17 or more legible words. CTH numbers follow the Portal CTH (Košak et al., 2026).

(Košak, 2026). The Ambazzi rituals (CTH 391, 429, 463) show Luwian influence but no stated link to Arzawa (Christiansen, 2006), so they stay outside the western group; Section 5.1 tests that decision.

Two sets leave the reference and are placed like orphans: the texts of CTH 390–469 that no defining study claims, and CTH 500, "fragments of festival and incantation rituals from Kizzuwatna", which is a running list of fragments rather than a composition. For CTH 701, the subgroups a–f of Salvini and Wegner (1986) and the script dates come from the Konkordanz and serve as partial ground truth.

4 Method

[Next draft: features, assignment, margin and abstention, word categories and their audit; Figure 1, the worked example.]

5 Results

[Next drafts.]

	analysed only		all manuscripts	
Tradition	n	acc.	n	acc.
Arzawa (western) rituals	11	55%	35	83%
Hattian cult layer	479	95%	527	93%
Kizzuwatna rituals	165	92%	234	92%
Kizzuwatna and Lower Land (Hurr.)	318	82%	323	79%
Ištanuwa and Lallupiya festivals	52	83%	55	78%
Luwian rituals (Lower Land)	10	40%	106	86%
Mixed (Hurr., Syr., Luw.)	200	67%	206	68%
Palaic festivals	34	82%	36	81%
Syro-Mesopotamia (Hurr.)	167	67%	175	67%
Ḫuwaššanna festivals	189	90%	200	88%
All	1,625	84%	1,897	84%

Table 2: Spellings only: leave-one-out accuracy per tradition in the sign space, with the reference limited to manuscripts that have word analyses, and with all manuscripts of 17 or more legible words.

5.1 Testing scholarly groupings

[Composition recovery and the Ambazzi rituals.]

5.2 Spellings only, with every manuscript

The gap in the annotation falls on the smallest traditions. We therefore repeat the leave-one-out evaluation in the sign space alone, where every legible word counts, once with the analysed manuscripts only and once with all manuscripts of 17 or more legible words (Table 2). The reference grows from 1,625 to 1,897 manuscripts, and overall accuracy stays at 84%. The Luwian rituals grow from 10 to 106 manuscripts and their accuracy from 40% to 86%; the western rituals from 11 to 35 (55% to 83%). The other traditions change little. Of the 847 orphans the main model places, 635 are also placed with the full reference and 608 of these keep their tradition; 37 move to the Luwian rituals and 16 to the western rituals. The worked example is affected: with the full reference KBo 41.22 still points first to the Luwian rituals, but its lead (0.010) falls below the margin, and it is no longer placed. Placements in the two small ritual traditions should be read with this in mind.

5.3 What a placement rests on

[Decision split by word category and layer.]

6 Discussion and conclusion

[Genre before origin; QTM in one paragraph.]

Limitations

[Uneven editions, small traditions, the unchecked gloss list.]

References

Daliah Bawanypeck. 2005. *Die Rituale der Auguren*. Number 25 in Texte der Hethiter. Winter, Heidelberg.

John Burrows. 2002. 'Delta': A measure of stylistic difference and a guide to likely authorship. *Literary and Linguistic Computing*, 17(3):267–287.

Birgit Christiansen. 2006. *Die Ritualtradition der Ambazzi: Eine philologische Bearbeitung und entstehungsgeschichtliche Analyse der Ritualtexte CTH 391, CTH 429 und CTH 463*. Number 48 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Maciej Eder. 2015. Does size matter? authorship attribution, small samples, big problem. *Digital Scholarship in the Humanities*, 30(2):167–182.

Shai Gordin. 2015. *Hittite Scribal Circles: Scholarly Tradition and Writing Habits*. Number 59 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Shai Gordin, Gai Gutherz, Ariel Elazary, Avital Romach, Enrique Jiménez, Jonathan Berant, and Yoram Cohen. 2020. Reading Akkadian cuneiform using natural language processing. *PLOS ONE*, 15(10):e0240511.

Shai Gordin, Emma Yavasan, and Avital Romach. 2025. Matters of genre, style and orthography: Hittite scribal habits through the stylometric lens. In Susanne Görke, Adam Kryszeń, Gerfrid G. W. Müller, and Charles W. Steitler, editors, *The Tablets and Their Scribes: The Palaeography of Hittite Cuneiform and Writing Practices in Hittite Anatolia*, number 73 in Studien zu den Boğazköy-Texten, pages 107–123. Harrassowitz, Wiesbaden.

Susanne Görke. 2010. *Das Ritual der Aštu (CTH 490): Rekonstruktion und Tradition eines hurritisch-hethitischen Rituals aus Boğazköy/Ḥattuša*. Number 40 in Culture and History of the Ancient Near East. Brill, Leiden.

Susanne Görke. 2020. Ḥatti, Pala und Luwiya: Ausgewählte palaische und luwische festrituale im vergleich. In Susanne Görke and Charles W. Steitler, editors, *Cult, Temple, Sacred Spaces: Cult Practices and Cult Spaces in Hittite Anatolia and Neighbouring Cultures*, number 66 in Studien zu den Boğazköy-Texten, pages 291–304. Harrassowitz, Wiesbaden.

Susanne Görke. 2022. Hurrian and Luwian elements in the Kizzuwatna religious texts. *Altorientalische Forschungen*, 49:148–157.

Manfred Hutter. 2003. Aspects of Luwian religion. In H. Craig Melchert, editor, *The Luwians*, Handbuch der Orientalistik I/68, pages 211–280. Brill, Leiden.

Tommi Jauhiainen, Heidi Jauhiainen, Tero Alstola, and Krister Lindén. 2019. Language and dialect identification of cuneiform texts. In *Proceedings of the Sixth Workshop on NLP for Similar Languages, Varieties and Dialects*, pages 89–98, Ann Arbor, Michigan. Association for Computational Linguistics.

Patrick Juola. 2008. Authorship attribution. *Foundations and Trends in Information Retrieval*, 1(3):233–334.

Jörg Klinger. 1996. *Untersuchungen zur Rekonstruktion der hattischen Kultschicht*. Number 37 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Moshe Koppel, Jonathan Schler, and Shlomo Argamon. 2011. Authorship attribution in the wild. *Language Resources and Evaluation*, 45(1):83–94.

Silvin Košak. 2026. Konkordanz der hethitischen keilschrifttafeln. hethiter.net/: hetkonk, `https://hethport.net/hetkonk/`. CTH 701 page accessed 2026-10-16.

Silvin Košak, Gerfrid G. W. Müller, Susanne Görke, and Charles W. Steitler. 2026. Catalog der texte der Hethiter (CTH). hethiter.net/: CTH, `https://hethport.net/CTH/`. Accessed 2026-10-16.

Emmanuel Laroche. 1971. *Catalogue des textes hittites*. Klincksieck, Paris.

Jared L. Miller. 2004. *Studies in the Origins, Development and Interpretation of the Kizzuwatna Rituals*. Number 46 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Gerfrid Müller, Doris Prechel, Elisabeth Rieken, and Daniel Schwemer. 2026. Thesaurus linguarum hethaeorum digitalis (TLHdig) beta version 0.3. Zenodo.

Elisabeth Rieken and Daniel Schwemer. 2026. Corpus der hethitischen festrituale (HFR). Akademie der Wissenschaften und der Literatur Mainz, hethiter.net/: HFR, `https://hethport.net/HFR/`. Accessed 2026-10-17.

Mirjo Salvini and Ilse Wegner. 1986. *Die Rituale des AZU-Priesters*. Corpus der hurritischen Sprachdenkmäler I/2. Multigrafica, Roma.

Walter J. Scheirer, Anderson de Rezende Rocha, Archana Sapkota, and Terrance E. Boult. 2013. Toward open set recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(7):1757–1772.

Fabian Simonjetz, Jussi Laasonen, Yunus Cobanoglu, Alexander Fraser, and Enrique Jiménez. 2024. Reconstruction of cuneiform literary texts as text matching. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 13712–13721, Torino, Italia. ELRA and ICCL.

Efstathios Stamatatos. 2009. A survey of modern authorship attribution methods. *Journal of the American Society for Information Science and Technology*, 60(3):538–556.

Rita Strauß. 2006. *Reinigungsrituale aus Kizzuwatna: Ein Beitrag zur Erforschung hethitischer Ritualtradition und Kulturgeschichte*. de Gruyter, Berlin and New York.

Andrea Trameri. 2025. *Kizzuwatna: History of Cilicia in the Middle and Late Bronze Age (ca. 2000–1200 BC)*. Handbuch der Orientalistik I/185. Brill, Leiden.

Theo van den Hout. 2009. A century of Hittite text dating and the origins of the Hittite cuneiform script. *Incontri Linguistici*, 32:11–35.

Willemijn Waal. 2015. *Hittite Diplomatics: Studies in Ancient Document Format and Record Management*. Number 57 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Mark Weeden. 2011. *Hittite Logograms and Hittite Scholarship*. Number 54 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Ilse Wegner and Mirjo Salvini. 1991. *Die hethitisch-hurritischen Ritualtafeln des (ḥ)išuwa-Festes*. Corpus der hurritischen Sprachdenkmäler I/4. Multigrafica, Roma.

Ilya Yakubovich and Alice Mouton. 2023. *Luwili: Hittite-Luwian Ritual Texts Attributed to Puriyanni, Kuwattalla and Šilalluḥi (CTH 758–763). Volume II: Discussion and Glossary*. Number 72,2 in Studien zu den Boğazköy-Texten. Harrassowitz, Wiesbaden.

Emma Yavasan and Shai Gordin. 2025. From clay to code: Transforming Hittite texts for machine learning. In *Proceedings of the Second Workshop on Ancient Language Processing (ALP 2025)*, pages 77–86. Association for Computational Linguistics.

Vasiliy Yugay, Kartik Paliwal, Yunus Cobanoglu, Luis Sáenz, Ekaterine Gogokhia, Shai Gordin, and Enrique Jiménez. 2024. Stylistic classification of cuneiform signs using convolutional neural networks. *it – Information Technology*, 66(1):15–27.